

Technical Talking Points: Specifications, Equations, and Anticipated Questions

Companion to “The Methodology Revisions and Their Results” — Stack v0.9.67 — August 11, 2026

0. The one-slide version

1. Price discovery: fixed- β VECM, shares near parity unconditionally (CS_{ES} 0.38 at 1s, 0.46 at 10ms); volatile-minus-benchmark gap +0.20/+0.12, permutation $p = 0.048/0.047$.
2. Tandem: marginal-preserving null keeps Table 5 (log-OR $1.10 \rightarrow 1.39 \rightarrow 1.47$, monotone); frequency-matched null kills the action-time comparison ($33.5\times \rightarrow 1.32\times \rightarrow 0.97\times$); OFI-innovation correlation $0.742 \rightarrow 0.835$.
3. Mechanism: interacted ECM-SDE, δ -term $t = 5.24$ (mid)/11.70 (microprice) at 10ms vs 0.25/1.35 at 1s; stressed half-life ≈ 3 min.
4. Table 9: rolling-window MA(W) artifact demonstrated by the 15-vs-60 lag experiment; RealBar (window-free) keeps the same three Romano–Wolf-significant cells at both depths.
5. Inference: pooled-iid $t = 253 \rightarrow$ day-clustered 4.62 (wild-cluster $p = 0.001$); Rigobon pre-test fails (variance-ratio spread $0.055 < 0.15$) $\rightarrow$ Cholesky bracket reported.

1. Sample and data engineering

Selection and pairing. Volatile days = the ten largest SPY daily ranges (high–low)/close in 2022–2026, ranked from daily OHLC. Each is paired to the same weekday 350–371 calendar days earlier (target 364 = exactly 52 weeks), which controls weekday seasonality in volume and liquidity while staying outside the crisis episode. The ranking is re-derivable by anyone from a public daily file (`rank_sample.py --csv spy_daily.csv --compare` prints the ranking, the suggested pairs, and where each shipped day ranks under the user’s own data); `validate_sample.py` machine-checks the pairing rules.

Q: Why intraday range rather than realized volatility for selection? Range is computable from public daily OHLC — vendor-independent and referee-reproducible — and is monotone in RV for these sessions. The selection criterion must not depend on the microstructure data being analyzed.

Q: Why a 350–371-day window instead of exactly 364? Holiday and closure adjustments; the window is the tightest that leaves every volatile day pairable on the same weekday.

Two grids, one pull. Extraction happens once, at 10ms (full ladder, both legs). The 1s frames are *derived*: last 10ms snapshot of each second for state variables (prices, depth, spreads), within-second sums for flow variables (OFI, volume), with all-missing seconds left as NaN rather than zero-filled. Automated gate tests verify derived-equals-directly-extracted.

Q: How do we know a 1s-vs-10ms difference is resolution and not data? Because there is no second dataset. Any coarse/fine difference is the aggregation operator by construction.

ES leg. Front-month ES rebuilt from the CME venue ladder (not the consolidated feed), contract chosen by a calendar roll rule, with an independent message-feed replay cross-checked against the ladder day by day. One open disagreement (2024-12-18, ESH5) is quarantined until diagnosed.

Halt masking. Halt windows (900s per MWCB session) and reopen seams are set to NaN at the frame level; every estimator drops non-finite *rows of its own design matrix*. Crucially, prices are differenced on the raw grid *before* dropping — compressing NaNs first would splice the last pre-halt price to the reopen price and manufacture exactly the pseudo-return the mask exists to remove.

Q: Doesn't masking throw away the most interesting data? No trading occurs inside a halt; the mask removes a non-observation, not information. All stressed *trading* stays: the four 2020 sessions still contribute $\approx 22.5k$ (1 s)/2.25M (10ms) tradable observations each.

2. Price discovery: the VECM, CS, IS, and the regime split

The system. For log mids $p = (p_{SPY}, p_{ES})'$:

$$\Delta p_t = \alpha \cdot z_{t-1} + \sum_{i=1..k} \Gamma_i \Delta p_{t-i} + \varepsilon_t, \quad z_t = p_{SPY,t} - p_{ES,t} \text{ (demeaned)}$$

with the cointegrating vector *fixed* at $\beta = (1, -1)$ and per-day `ec_valid` screening (the error-correction term must actually equilibrate: correct loading signs and a stationary z).

Q: Why fix β instead of estimating it? The two legs are claims on the same index; the arbitrage relation is $(1, -1)$. Estimating β on intraday data lets microstructure noise rotate the cointegrating vector, and every downstream share (CS, IS) inherits that rotation. Fixing β is a restriction that is true by construction, and `ec_valid` verifies it day by day (0 of 24 days fail, at both grids).

Component share (Gonzalo–Granger). From the loadings $\alpha = (\alpha_{SPY}, \alpha_{ES})'$, the common permanent component weights are $\alpha_{\perp}$ (the direction orthogonal to adjustment):

$$CS_{ES} = |\alpha_{SPY}| / (|\alpha_{SPY}| + |\alpha_{ES}|) \quad (\text{with equilibrating signs})$$

— the leader is the leg that does *not* adjust. No ordering assumption enters. Full-panel means: 0.384 (1s), 0.455 (10ms).

Information share (Hasbrouck). From the VMA long-run impact row ψ and residual covariance Ω , with $F = \text{chol}(\Omega)$:

$$IS_j = ([\psi F]_j)^2 / (\psi \Omega \psi')$$

computed under both Cholesky orderings $\rightarrow$ [lower, upper] bounds; the midpoint is quoted when one number is needed. Means: 0.484 (1s), 0.542 (10ms).

Q: Why are the 1s bounds so wide (e.g., [0.004, 0.861] on 2023-08-07)? At 1s the residual cross-correlation is near one — nearly all adjustment is within-interval — so the ordering assumption performs the entire attribution. The bounds are the estimator honestly reporting that. At 10ms the same day gives [0.244, 0.523]: genuine asynchrony breaks the simultaneity and the data, not the ordering, do the attribution. This is the technical argument for quoting IS at 10ms.

The regime split and its inference. Day-level CS_{ES} means: 0.269 $\rightarrow$ 0.466 (1s), 0.387 $\rightarrow$ 0.504 (10ms). Two tests:

1. *Day-level permutation*: regime labels permuted across days, statistic = difference in regime means; exact under exchangeability; $p = 0.048/0.047$.
2. *Pooled FE panel VECM*: lags built strictly within-day (no overnight seam-crossing), day fixed effects, day-clustered SEs, volatile $\times$ EC interaction; $t = -1.50$ (SPY), -1.60 (ES), $p \approx 0.13$ – 0.15 .

Q: Why lead with permutation rather than the panel t ? With $G = 24$ clusters the clustered t relies on asymptotics in G ; the permutation test is exactly sized at any N . Both are reported; they agree in sign, and the appropriately sized one rejects.

Q: Did the headline change because of the method or the sample? Both, and the memo says which: the halt fix and the venue-ladder ES leg are method (they mechanically reduce measured futures dominance on MWCB days); the 2022–2026 panel is sample. The *regime* result — futures take over under stress — holds on the new sample at both grids and is the claim the revision should make.

Jumps and co-jumps. Lee–Mykland classifier: $L_i = r_i/\hat{\sigma}_i$ with $\hat{\sigma}$ a local bipower (jump-robust) volatility; threshold from the extreme-value limit of $\max|L|$ under the null. At 10ms, $\approx 58\%$ of common-factor quadratic variation is discontinuous. IS recomputed separately on jump and continuous components: $IS_{j,ES} = 0.563$ vs $IS_{c,ES} = 0.390$. Co-jump timing on 2020-03-09: of 24,346 common jumps, ES first 7,610, SPY first 1,669 (4.6:1), 15,067 simultaneous at 10ms, sign agreement 94.8%.

Q: Isn't jump detection threshold-sensitive? The 1s comparison is reported both ways (truncation 17%, LM 6%) precisely to show the sensitivity; the 10ms LM classifier is the one operating in its intended asymptotic regime. The co-jump lead ratio is a count, not a model output — it needs no orthogonalization and no threshold beyond jump flagging.

3. Tandem order flow: the two corrected nulls

Table 5 (marginal-preserving null). Per bar, sign each leg's net order flow; build the 2×2 sign table. The published null was $\text{Binomial}(n, \frac{1}{2})^2$ — fair coins *and* independence. The corrected null is independence *conditional on the observed marginals*: expected corner mass = $\hat{p}_+^{\text{SPY}} \cdot \hat{p}_+^{\text{ES}}$ (and analogues), estimated per day. Reported statistics:

$$\log\text{-OR} = \log(n_{11}n_{00}/n_{10}n_{01}); \quad \text{corner asymmetry} = \text{local log-odds(sell)} - \text{local log-odds(buy)}$$

Results: observed/null $1.27 \rightarrow 1.34 \rightarrow 1.37$ across baseline/volatile/MWCB; log-OR $1.098 \rightarrow 1.385 \rightarrow 1.470$ (ex-SSR 1.413); corner asymmetry ≈ 0 in every panel (no Rule-201 fingerprint — the dependence is symmetric two-way coordination). Pooled Woolf z -statistics exceed 60 everywhere; the quotable inference is day-clustered.

Q: Doesn't conditioning on marginals throw away signal? It removes exactly the component a referee will attribute to common one-sided pressure without cross-market coordination. What survives is the paper's actual claim. That it survives *and* is monotone in stress is a stronger exhibit than the old table.

Table 7 (frequency-matched null). The null corner probability is a function of the per-bar order counts, which collapse with bar size (mean orders per bar: 505/112 at 1s, 5.05/1.12 at 10ms, 1/1 in action time). Recomputing the null at each aggregation's actual counts: observed vs null = 11.9% vs 0.35% (33.5 $\times$) at 1s; 30.1% vs 22.9% (1.32 $\times$) at 10ms; 48.4% vs 50.1% (0.97 $\times$) in action time.

Q: So is the *fine-grid evidence* gone? The *level* evidence at action time is gone — it was benchmark arithmetic. The correct statement: coordination operates at the hundreds-of-milliseconds-to-seconds horizon and attenuates to chance at the arrival scale. That is economically sensible (latency, hedging horizon) and defensible.

Innovation-level correlation. Each leg’s OFI is prefiltered by its own past (AR/VAR prefilter per day); the residual cross-correlation is 0.742 (benchmark) vs 0.835 (volatile). This is the title phenomenon measured without any null construction.

GFEVD. The orthogonal FEVD (diagonal- Σ assumption) attributes 95% of ES return variance to ES flow. The Pesaran–Shin generalized decomposition evaluated at the measured innovation covariance,

$$\theta_{ij}(H) = \sigma_{jj}^{-1} \Sigma_h (e_i' A_h \Sigma e_j)^2 / \Sigma_h (e_i' A_h \Sigma A_h' e_i),$$

attributes 65% ES / 35–43% SPY, with shares not summing to one because common flow is counted toward both shocks — quoted explicitly as an upper bound on separability.

4. The mechanism: interacted ECM-SDE and cross-impact

Specification. With $s_t = \text{standardized log}(\text{total ES depth}) - \text{log}(\text{total SPY depth})$:

$$\Delta p_t = c + (\alpha + \delta \cdot s_t) \cdot z_{t-1} + \sum_{i=1..k} \Gamma_i \Delta p_{t-i} + \epsilon_t$$

Pooled across days with day fixed effects; δ is the mechanism (“arbitrage weakens as the book thins”). Lag order is selected on the *base* VECM — the EC $\times$ state interaction is p-invariant, so lag choice cannot tune the headline term. The implied adjustment speed $\kappa(s)$ gives half-life $\ln 2 / \kappa(s)$ at quantiles of s .

Results: δ -term $t = 0.25$ (mid)/1.35 (microprice) at 1s; 5.24/11.70 at 10ms; stressed half-life ≈ 3 minutes at 10ms; median IS_{ES} along the liquidity curve 0.67/0.74.

Q: Why did the day-by-day quantile splits disappear? They are underpowered and flip sign day to day; the pooled interaction estimates one parameter with day FE absorbing levels. The splits were a discretized, noisy version of the same regression.

Q: Why does 1s hide the effect? The book-state innovations that move κ live at sub-second horizons; averaging to 1s attenuates the regressor exactly where it varies.

Cross-impact. Return-on-flow regressions (own + cross): the cross coefficient roughly triples benchmark $\rightarrow$ volatile within each grid (10ms: 0.020 $\rightarrow$ 0.060; 1s: 0.198 $\rightarrow$ 0.268); on MWCB days ES-flow $\rightarrow$ SPY-return (0.13–0.25) is 2–5 $\times$ SPY-flow $\rightarrow$ ES-return (0.02–0.06).

Book state vs quoted spread. R^2 for |SPY returns|: 0.318 (book state) vs 0.138 (quoted spread); partial R^2 of state given spread = 0.179. The state variable is not a repackaged spread.

5. Table 9: the correlation system and the lag experiment

Design. FE panel VAR(p) on [Δ correlation-measure, Spread, WtdSpread, OFI, RV, MicroDev $\times$ both legs], lags within-day, day FE, Cholesky identification (futures block first), one- σ orthogonalized impact responses $\times 100$, day-cluster bootstrap SEs, Romano–Wolf joint stars.

The artifact, precisely. The published dependent variable is the change in a $W=100$ -bar rolling Pearson correlation. Adjacent windows share $W-1$ bars, so $\Delta\rho_t$ is a moving-average object of order $\approx W$ in the underlying returns: every shock echoes for exactly W bars. A lag polynomial with $p < W$ cannot whiten it; an information criterion therefore buys lags all the way to its bound (the earlier run’s BIC chose $p^* = p_{\max}$ — the mechanical restatement of the paper’s footnote 17).

The experiment. Identical data, identical estimator, $p = 15$ vs $p = 60$:

- Pearson/HY columns: impact responses shrink 3–10 $\times$ (several to zero at reported precision); of ten Romano–Wolf-starred cells, two survive at both depths; seven starred at 15 lose stars at 60; one appears only at 60. HY shrinks in parallel (it shares the window).
- The HY–Pearson delta (the Epps-artifact share of the published design) is itself lag-dependent: $\approx 74\%$ of the response at $p = 15$ (RV_{ES} benchmark: -0.380 of 0.517), $\approx 23\%$ at $p = 60$.
- RealBar column (Δ Fisher- z of non-overlapping 60s-bar realized correlation): magnitudes scale 3–7 $\times$ with SEs moving proportionally — one- σ impact responses are denominated in innovation size, which the lag depth re-sizes — but the *same three cells* are starred at both depths ($WtdSpread_{ES}/volatile$, $OFI_{ES}/benchmark$, $RV_{ES}/benchmark$), no starred cell changes sign, no cell is significant with opposite signs.

Q: Why not just run $p \geq W = 100$? It concedes the point: the model would be spending 100 lags modeling the measurement window. The fix is a dependent variable with no window. RealBar’s only residual structure is an MA(1) from adjacent bars sharing one noisy level estimate — bounded and at lag 1, unlike the W -lag artifact.

Q: Why 60s bars? Bias–variance: enough ticks per bar for a stable per-bar correlation estimate, short enough to leave ≈ 390 bars/day of resolution. The Fisher z (arctanh) transform stabilizes the variance of the per-bar estimates.

Q: If RealBar magnitudes aren’t stable either, what is Table 9 worth? Signs and joint significance — which are stable — plus the DCC corroboration (its starred cells: weighted-spread rows and $RV_{SPY}/benchmark$). The recommendation is explicitly a signs-and-significance exhibit with p fixed ex ante on the bar grid, where the criterion is not chasing a window.

Q: Why is DCC only corroboration? GARCH(1,1) per leg with a cDCC recursion ($Q_t = (1-a-b)\bar{Q} + a \cdot u_{t-1}u'_{t-1} + b \cdot Q_{t-1}$); estimated persistence $a+b = 0.9999$ — near-integrated, so its level responses are hard to interpret even though its recursive (window-free) structure makes it lag-robust, which it is in the data.

What survives everywhere (the quotable core). RV_{ES} is the only shock starred in Pearson, HY, and RealBar in both runs (volatility raises correlation); weighted-spread shocks are the only other jointly robust family, ES-side positive/volatile and negative/benchmark in every starred cell, SPY-side positive where starred; OFI_{ES} lowers benchmark-day correlation in the window-free column at both depths; MicroDev is null everywhere.

6. Inference and identification

Clustering. All pooled regressions: cluster-robust at the day level ($G = 24$). Flagship example: SPY-return $\sim$ ES-flow, pooled-iid $t = 253.3 \rightarrow$ clustered $t = 4.62$.

Wild-cluster bootstrap. Rademacher weights at $G = 24$; Webb six-point weights where a subsample has very few clusters (MWCB $G = 4$, where Rademacher yields only 2^4 distinct resamples). Flagship $p = 0.001$.

Permutation tests. Regime-label permutation at the day level for all volatile-vs-benchmark contrasts; exact under exchangeability; the headline evidence at $N = 24$.

Romano–Wolf. Stepdown FWER control across all cells of each multi-cell table (the joint stars in the Table 9 exhibits); degenerate cells (undefined t) are excluded from the stepdown and reported unstarred.

Rigobon het-ID. Identification through heteroskedasticity requires the regime shift to move the two legs' variances by different proportions. Pre-test: the cross-regime variance-ratio spread is 0.055 against a 0.15 threshold (and the relative eigengap condition also fails) — the stress regime scales both legs nearly equally, so the rotation is unidentified and het-ID point estimates are noise. With the MWCB days as a third regime, the over-identification statistic is available as a specification test. The draft should report the verdict row and the two-Cholesky bracket, not het-ID coefficients.

Q: Where does the 0.15 threshold come from? Calibration on simulated scale-vs-rotation regimes; the qualitative point is robust — at a 0.055 spread the identifying variation is an order of magnitude below where the estimator stabilizes, and the over-ID test is available as the formal check.

7. Rapid-fire: cross-cutting questions

“Is this reproducible?” One driver script replicates the pipeline end to end; a gate suite (compile checks, estimator unit tests, frozen golden-number regressions, driver-text invariants) runs before any stage; CI runs the full suite on every push. The sample selection is re-derivable from public daily data.

“Which published numbers change and why?” Three causes, cleanly separable: (i) halt masking (MWCB-day estimates), (ii) the corrected nulls (Tables 5/7), (iii) the window-free correlation measures (Table 9). Each is documented with the before/after in the corresponding memo section.

“What is genuinely new versus corrected?” New: the 10ms grid and everything only it can see (tight IS bounds, the ECM-SDE result, jump/co-jump leadership, the innovation correlation, GFEVD). Corrected: nulls, inference, halt handling, Table 9's dependent variable.

“What's still open?” The ESH5 2024-12-18 replay-vs-ladder disagreement (quarantined), and the 2025-06-13 contract-month decision (the calendar rule picked the 13.3%-volume contract; re-extract pinned to the majority contract, or keep the rule and report the share).

“Biggest referee risk?” The unconditional-headline retreat. The preemptive answer: the regime-conditional claim is stronger evidence *for the paper's mechanism* than the unconditional claim ever was — leadership appears exactly when tandem trading under extraordinary volatility is operating, at both measurement frequencies, by an exactly sized test.